A Citation Index for Federal Regulations

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July 2, 2026

Abstract

Federal regulations are required to rest on reasoned decision-making and the best available science, yet little systematic evidence exists on the research actually cited in rulemaking. We construct a comprehensive, instance-level citation index for federal regulations by extracting and cataloging academic citations from the full text of 113,712 final rules published in the Federal Register between 1995 and early 2025. Using large language models to parse regulatory text and CrossRef and OpenAlex to enrich bibliographic metadata, we identify 85,327 citation instances spanning 55,520 unique papers by 72,308 unique authors. Citations to academic research grew roughly eightfold over the period, from about 1,100 per year in the mid-1990s to 8,778 in 2024, with the number of rules citing research and the number of citations per citing rule both rising substantially. Citation patterns vary widely across agencies in volume and disciplinary focus: the Department of Health and Human Services and the Environmental Protection Agency together account for nearly 40 percent of all citations. The dataset—a Google Scholar for rulemaking—enables new research on the role of evidence in policymaking.

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1 Introduction

The Administrative Procedure Act requires federal agencies to engage in "reasoned decision-making" when promulgating regulations, and under the hard-look doctrine of *Motor Vehicle Manufacturers Association v. State Farm*, 463 U.S. 29 (1983), courts review whether an agency's explanation is supported by the record before it. Executive orders dating back decades—from President Reagan's Executive Order 12291 to President Clinton's E.O. 12866 and its successors—mandate cost-benefit analysis grounded in evidence [Sunstein, 2013]. Recent decisions have only raised the stakes for the evidentiary record: in *Loper Bright Enterprises v. Raimondo*, 603 U.S. 369 (2024), the Supreme Court overturned *Chevron* deference, and in *Ohio v. EPA* (2024) the Court stayed a major rule on the ground that the agency had not adequately explained its decision or responded to comments raised during the rulemaking. The quality and breadth of the evidence agencies cite is increasingly a live question in judicial review.

Despite the centrality of evidence to regulatory legitimacy, the research actually cited in federal regulations has never been cataloged at scale. Legal scholars have long debated how science enters (and is sometimes camouflaged in) regulatory justifications [Wagner, 1995], and a policy literature evaluates the quality of regulatory analyses [Ellig and McLaughlin, 2012]. The work closest to ours is Desmarais and Hird [2014], who hand-coded the research cited in a sample of regulatory impact analyses, and Costa et al. [2016], who study how political attention affects science use in those documents; the commercial Overton database indexes research citations in policy documents worldwide [Szomszor and Adie, 2022]. What has been missing is an open, instance-level index covering the full text of every final rule over a long horizon—the regulatory analogue of a citation index for the academic literature.

This paper builds that index. We extract academic citations from the full text of 113,712 final rules published in the Federal Register from 1995 through early 2025, using a large language model to parse citations from regulatory text at a cost that would have been prohibitive only a few years ago [Dell, 2025]. Our dataset identifies 85,327 citation instances, which we resolve to 55,520 unique papers by 72,308 unique authors. Of the roughly 113,000 final rules in our sample, 7,002 (6.2 percent) contain at least one academic citation.

The resulting dataset functions like a Google Scholar for federal regulations: for each paper, we record how many times it was cited in regulations, by which agencies, and in which years; for each author, we aggregate regulatory citations across all their work; for each agency, we can identify the journals and researchers most frequently relied upon.

Three broad findings emerge. First, citations to academic research in federal regulations grew roughly eightfold over three decades, from about 1,100 per year in the mid-1990s to 8,778 in 2024. A simple decomposition shows the growth comes both from more rules citing research (the number of citing rules grew 3.3-fold) and from a deeper evidentiary record within citing rules (citations per citing rule grew 2.4-fold). Second, citation patterns differ sharply by agency, tracking each agency's statutory mandate: the Department of Health and Human Services (HHS) cites medical journals such as *JAMA* and the *New England Journal of Medicine*, the Environmental Protection Agency (EPA) relies on environmental health and atmospheric science research, and the Securities and Exchange Commission (SEC) draws heavily on finance and economics journals. Third, the evidence base is broad: regulations cite work from 5,691 distinct journals, and the most-cited journal accounts for only half of one percent of all citations.

2 Data and Methods

This section summarizes the construction of the index; Appendix B documents each stage in detail, and a replication package accompanies the paper.

2.1 Corpus

Our source corpus comprises the full text of every final rule published in the Federal Register from 1995 through January 2025—113,712 documents in total. We obtained document metadata (title, publication date, issuing agencies, document number) from the Federal Register API and downloaded the full text of each rule.

2.2 Citation Extraction

We extract academic citations in two stages. First, we identify footnotes in each document using the numbered-footnote convention of Federal Register typesetting, along with parenthetical citations in body text and formal reference sections. We pre-filter footnotes that contain only legal citations (to the Code of Federal Regulations, United States Code, court cases, and other legal authorities) using regular expressions, retaining footnotes that potentially reference academic or scientific literature.

Second, we use a large language model (Claude, Anthropic) to parse the remaining footnotes and extract structured citation information: author names, title, journal or publisher, year of publication, and DOI when available. To handle documents with large numbers of footnotes, we chunk footnotes into batches of 50 and process each batch independently through the Anthropic Message Batches API. Total extraction cost for the full corpus was approximately $550.

2.3 Enrichment and Disambiguation

We enrich extracted citations by querying the CrossRef API, matching on DOI, title, and author to retrieve DOIs, canonical journal names, ISSNs, and structured author records. After discarding unreliable matches (described below), 32.3 percent of citation occurrences (and 30 percent of unique papers) carry a DOI or CrossRef match.

Journal names are standardized against a registry built from the Web of Science citation indices (SCIE, ESCI, AHCI), Journal Citation Reports abbreviation tables, and manual correction files; the registry maps roughly 48,800 canonical journal titles and 37,300 ISSNs. This registry is a lookup resource; the number of distinct journals actually cited in regulations is 5,691.

Author names undergo a disambiguation pipeline that normalizes name variants, resolves bare last names using CrossRef structured data where available, and applies a Union-Find algorithm that merges two variants only with corroborating evidence—appearing on the same paper with compatible names, co-authorship overlap, or a CrossRef identity match. The pipeline consolidates 24,145 variant names, yielding 72,308 unique author identities.

Paper deduplication proceeds in two passes: exact grouping by DOI and normalized title–year combinations, followed by fuzzy title matching. An iterative enrichment step propagates metadata across matching citations within the corpus.

2.4 Correcting Systematic Matching Errors

Automated bibliographic matching produces systematic errors that we identify and correct. Three classes matter most. First, CrossRef sometimes attributes citations to the SSRN working-paper repository when a copy of the cited work was posted there; we resolve those to published journal versions via the OpenAlex API where a published version exists (500 of 1,075 unique SSRN papers), and revert the remainder to the original extracted citation. Second, CrossRef sometimes matches citations to database and platform entries—review databases, abstract services, and publisher eBook platforms—rather than the true source; we revert these as well. Third, low-similarity title matches on citations that the language model classified as government reports or other gray literature are unreliable—for example, OMB's Circular A-4 ("Regulatory Analysis") was matched to a biomedical article titled "Circular RNA: New Regulatory Molecules"—and we revert all such matches under a systematic rule. In total these corrections revert 5,421 citation instances (6.4 percent); each reverted record retains a flag and the discarded match, so the decisions are auditable. Appendix B.7 gives details.

2.5 Scope and Limitations

Four limitations should guide use of the data. First, the corpus is final-rule preamble text: citations that appear only in regulatory impact analyses, technical support documents, or docket materials are not captured, so our counts are a lower bound on the evidentiary record, and the paper measures citation in *final rules*, not everything in the rulemaking process. Second, extraction is footnote-centric; although we also capture in-text parentheticals and reference lists, agencies that cite outside these formats will be undercounted. Third, bibliographic matching is imperfect: we correct the systematic error classes above, but residual misattributions remain, and we have not yet completed a formal precision-recall audit of the extraction step (the replication package includes the sampling machinery for one). Fourth, author disambiguation is conservative: bare surnames ("Smith") and initial-only names that cannot be attributed to a single person are kept as separate identities and excluded from author-level rankings, so author statistics understate concentration among frequently-cited researchers. Citation counts measure what agencies cite, not the causal influence of that research on regulatory outcomes.

3 Results

3.1 Summary Statistics

Table 1 presents an overview of the dataset. From 113,712 final rules, we extract 85,327 citation instances from 7,002 regulations (6.2 percent). These resolve to 55,520 unique papers by 72,308 unique authors. Reports and gray literature constitute the largest share of citations (38.5 percent), followed by unclassified references (29.8 percent) and peer-reviewed journal articles (25.4 percent).

Table 1: Summary Statistics

Measure	**Count**
Final rules in corpus (1995–2025)	113,712
Rules with ≥ 1 academic citation	7,002 (6.2%)
Total citation instances	85,327
Unique papers (after deduplication)	55,520
Unique authors (after disambiguation)	72,308
Distinct journals cited	5,691
Citations with DOI or CrossRef match	27,540 (32.3%)
By citation type:	
Reports & gray literature	32,822 (38.5%)
Other / unclassified	25,413 (29.8%)
Journal articles	21,683 (25.4%)
Books	2,938 (3.4%)
Working papers	1,367 (1.6%)
Conference papers	1,040 (1.2%)
Dissertations	64 (0.1%)

Notes: A citation instance is one appearance of a citation in one final rule. Papers are unique cited works after deduplication; authors are unique identities after disambiguation. "Distinct journals cited" counts distinct journal titles observed in the citations (not the size of the journal-name reference registry). Citation types are assigned by the language model at extraction.

3.2 Citations Over Time

Figure 1 displays the number of academic citations in final rules by the rule's publication year. Citations grew roughly eightfold, from 1,108 in 1996 to 8,778 in 2024. The growth operates on both margins: the number of rules citing any research grew 3.3-fold between 1996 and 2024, and citations per citing rule grew 2.4-fold. Growth has been uneven, with spikes in years when major regulations were finalized—2016 (6,289 citations) and 2020 (6,436) stand out—and the series reaches its maximum in 2024.

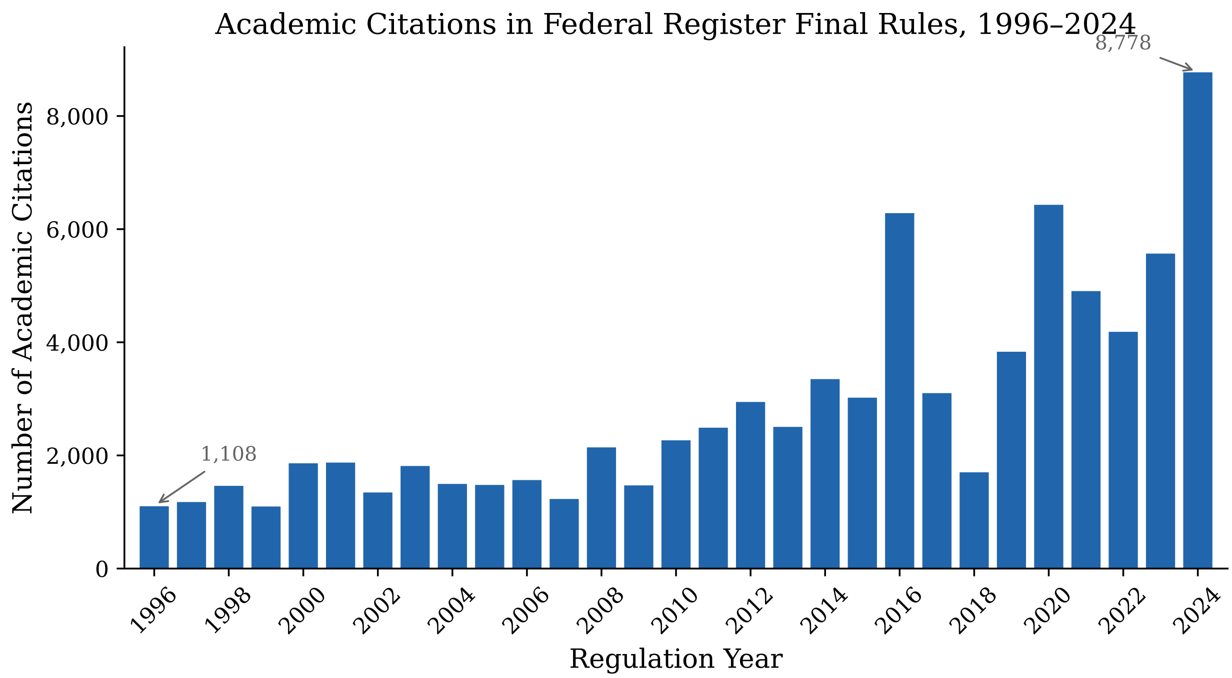

Figure 1: Academic citations in Federal Register final rules by regulation publication year, 1996–2024. The corpus covers 1995 through January 2025; 1995 (265 citations) and the partial year 2025 are omitted from the figure.

3.3 Citations by Agency

Citation volume varies widely across agencies. Table 2 reports the top 25 agencies by total citation count. HHS leads with 17,093 citations, followed by the EPA (15,743) and the Department of the Interior (12,142); these three account for over half of all citation instances. Volume tracks both regulatory output and the evidentiary intensity of each agency's docket:

much of the HHS total comes from Medicare payment and quality rules issued by CMS, Interior's from Fish and Wildlife Service species listings, and Commerce's from NOAA fisheries and protected-species rules.

Table 2: Top 25 Agencies by Total Academic Citations, 1995–2025

Rank	**Agency**	**Citations**
1	Health and Human Services Department	17,093
2	Environmental Protection Agency	15,743
3	Interior Department	12,142
4	Commerce Department	9,164
5	Labor Department	6,350
6	Securities and Exchange Commission	5,668
7	Transportation Department	4,739
8	Energy Department	3,487
9	Treasury Department	2,857
10	Agriculture Department	2,593
11	Consumer Financial Protection Bureau	2,118
12	Homeland Security Department	1,763
13	Education Department	1,594
14	Commodity Futures Trading Commission	1,012
15	Federal Communications Commission	927
16	Federal Reserve System	788
17	Justice Department	752
18	Federal Trade Commission	666
19	Housing and Urban Development Department	581
20	Federal Deposit Insurance Corporation	543
21	Consumer Product Safety Commission	485
22	Federal Housing Finance Agency	322
23	Personnel Management Office	294
24	Social Security Administration	253
25	Veterans Affairs Department	233

Notes: Counts are citation instances in rules listing the agency among the issuing agencies. A rule issued jointly by several agencies counts toward each, so the column is not additive. Sub-agencies whose rules always co-list a parent department (e.g., CMS under HHS, the Fish and Wildlife Service under Interior, NOAA under Commerce, the IRS under Treasury, FERC under Energy) are folded into the parent's total rather than listed separately.

3.4 Top Journals

Table 3 lists the 25 most-cited journals. Medical journals dominate the top of the list, driven by HHS rulemaking: *JAMA* ranks first, followed by the *New England Journal of Medicine* and *Health Affairs*. Finance journals—the *Journal of Financial Economics* and *The Journal of Finance*—rank fourth and fifth, reflecting SEC rulemaking and our resolution of SSRN working papers to their published venues. Environmental journals round out the top ten. The full top-100 list appears in Appendix Table 6.

Table 3: Top 25 Most-Cited Journals

Rank	**Journal**	**Citations**
1	Journal of the American Medical Association	430
2	New England Journal of Medicine	399
3	Health Affairs	371
4	Journal of Financial Economics	325
5	The Journal of Finance	320
6	Environmental Health Perspectives	267
7	Morbidity and Mortality Weekly Report	245
8	Environmental Science & Technology	189
9	SAE Technical Paper Series	176
10	Journal of the American Geriatrics Society	170
11	American Economic Review	166
12	Annals of Internal Medicine	162
13	Energy Policy	139
14	Medical Care	130
15	Journal of General Internal Medicine	130
16	Quarterly Journal of Economics	128
17	Review of Financial Studies	125
18	Atmospheric Environment	122
19	JAMA Network Open	121
20	Proceedings of the National Academy of Sciences	114
21	Health Services Research	113
22	The Lancet	107
23	American Journal of Public Health	101
24	Archives of Physical Medicine and Rehabilitation	94
25	British Medical Journal	93

Notes: Counts are citation instances, 1995–2025. Journal names are standardized to canonical full titles; abbreviated forms in Table 4 and Appendix Table 6 refer to the same standardized identities.

3.5 Disciplinary Specialization Across Agencies

The citation patterns reveal strong disciplinary specialization. Table 4 shows the top five journals cited by twelve major regulatory agencies. HHS relies on clinical medicine journals. The EPA draws on environmental health and atmospheric science. The SEC's top journals are the leading finance and accounting outlets. The Department of Labor's citations span health policy and general economics. Interior and Commerce cite ecology and wildlife journals, consistent with Fish and Wildlife Service and NOAA rulemaking driving those totals.

Table 4: Top 5 Journals by Agency

Agency	**Journal**	**Cites**	**Agency**	**Journal**	**Cites**
	N. Engl. J. Med.	358		J. Financial Economics	288
	JAMA	347		The Journal of Finance	249
HHS	Health Affairs	338	*SEC*	J. Accounting & Econ.	88
	MMWR	173		The Accounting Review	79
	J. American Geriatrics Society	170		Review of Financial Studies	72
	Environ. Health Persp.	251		Health Affairs	68
	Environ. Sci. & Tech.	172		American Economic Review	44
EPA	Atmospheric Environment	122	*Labor*	Q. J. Economics	39
	SAE Technical Paper Series	96		N. Engl. J. Med.	39
	J. Air & Waste Mgmt. Assoc.	64		MMWR	28
	J. Wildlife Management	23		Marine Mammal Science	20
	Journal of Wildlife Diseases	10		Endangered Species Research	11
Interior	Poultry Science	7	*Commerce*	Fisheries Research	10
	Freshwater Biology	7		Copeia	7
	Environmental Pollution	6		Pathology	6
	SAE Technical Paper Series	124		Energy Policy	88
	Environ. Health Persp.	37		Review of Economic Studies	30
Transportation	Energy Policy	33	*Energy*	J. Geophys. Res.: Atmospheres	22
	Rev. Econ. and Statistics	26		AIP Conference Proceedings	21
	Transp. Res. Record: J. Transp. Res. Board	25		Climate Policy	18
	Health Affairs	63		Journal of Food Protection	50
	N. Engl. J. Med.	25		Am. J. Agricultural Econ.	28
Treasury	American Economic Review	22	*Agriculture*	Poultry Science	21
	Finance and Econ. Discussion Series	21		Am. J. Clinical Nutrition	19
	Journal of Health Economics	19		Agricultural Statistics	17
	J. Economic Perspectives	22		Federal Reserve Bulletin	17
	Violence Against Women	19		Finance and Econ. Discussion Series	17
Education	Econ. of Education Review	18	*CFPB*	American Economic Review	16
	American Economic Journal	16		Q. J. Economics	16
	Student Loans and the Dynamics of Debt	13		Supreme Court Econ. Review	16

Notes: Counts are citation instances in rules listing the agency among the issuing agencies; sub-agency citations are included in the parent department's column (e.g., CMS in HHS, FERC in Energy). The twelve agencies are the highest-volume departments and independent agencies with distinct disciplinary profiles. Journal titles are abbreviated; full canonical titles appear in Table 3 and Appendix Table 6. Some frequently cited venues are not peer-reviewed journals (e.g., the Federal Reserve's Finance and Economics Discussion Series; *Agricultural Statistics*).

3.6 Top-Cited Authors in HHS Regulations

Table 5 reports the 25 most-cited authors in HHS regulations, along with each author's most-cited journal within those citations. The list is dominated by health services and outcomes researchers whose work informs Medicare quality measurement, hospital readmission policy, and payment reform. Karen Joynt Maddox (Washington University) ranks first with 165 citations, primarily for studies of social risk factors and hospital performance; Harlan Krumholz (Yale) and Arnold Epstein (Harvard) follow. The prevalence of *Circulation: Cardiovascular Quality and Outcomes* in the third column reflects CMS's reliance on the cardiovascular outcome-measures literature in hospital quality rulemaking.

4 Discussion

The dominance of reports and gray literature (38.5 percent of citations) alongside peer-reviewed journal articles (25.4 percent) reflects the practical nature of regulatory evidence. Agencies frequently cite government reports, technical documents, and industry studies that provide the applied inputs cost-benefit analysis requires—dose-response functions, market data, compliance cost surveys—and the peer-reviewed literature enters alongside these sources rather than instead of them.

The disciplinary composition of citations tracks each agency's statutory mandate: CMS rules cite clinical outcomes research, EPA rules cite environmental epidemiology and toxicology, SEC rules cite financial economics. The pattern is consistent with agencies drawing on domain-appropriate research rather than a generic evidence base, though citation counts alone cannot establish how that research shaped decisions.

The timing of working-paper citations offers a sharper observation. Among the 770 citation instances we resolve from SSRN postings to published journal articles, 88 percent cite research that had already appeared in a journal by the time the rule was published—regulators (or their commenters) were working from the freely available working-paper ver-

Table 5: Top 25 Most-Cited Authors in HHS Regulations

	Author	Cites	Most-cited journal
1	Karen E. Joynt	165	Circulation: Cardiovascular Quality and Outcomes
2	Harlan M. Krumholz	117	British Medical Journal
3	Arnold M. Epstein	88	Circulation: Cardiovascular Quality and Outcomes
4	Marc Elliott	80	Health Services Research
5	Rachael Zuckerman	79	Circulation: Cardiovascular Quality and Outcomes
6	Ashish K. Jha	62	JAMA
7	Alan M. Zaslavsky	58	Health Services Research
8	Vincent Mor	56	Health Affairs
9	David Bates	52	Annals of Internal Medicine
10	Sharon-Lise T. Normand	49	Circulation: Cardiovascular Quality and Outcomes
11	Peter K. Lindenauer	47	British Medical Journal
12	Susannah M. Bernheim	46	Circulation: Cardiovascular Quality and Outcomes
13	Bruce Landon	46	N. Engl. J. Med.
14	Glenn M. Chertow	44	Hemodialysis International
15	Kenneth J. Ottenbacher	44	Archives of Physical Med. and Rehabilitation
16	Amal N. Trivedi	43	N. Engl. J. Med.
17	David C. Grabowski	42	Health Affairs
18	Zhenqiu Lin	40	Circulation: Cardiovascular Quality and Outcomes
19	Amy Finkelstein	40	Health Affairs
20	Michael B. Rothberg	40	British Medical Journal
21	Amelia M. Haviland	40	Health Services Research
22	Joseph Newhouse	39	Quarterly Journal of Economics
23	Lisa G. Suter	39	J. General Internal Medicine
24	Dale W. Bratzler	38	N. Engl. J. Med.
25	Katrin Hambarsoomian	37	Medical Care Res. and Review

Notes: "Cites" counts citation instances in HHS-issued rules to papers on which the author appears, 1995–2025. "Most-cited journal" is the journal accounting for the most of those instances, computed from the index itself. Author identities come from the disambiguation pipeline described in Appendix B.6; bare-surname and initial-only identities that cannot be attributed to a single person are excluded. Karen Joynt Maddox published as Karen E. Joynt before 2017; the index merges both names into a single identity.

sion of published research. The remaining 12 percent cite research before it appeared in print. At the SEC, which accounts for most SSRN citations, 56 percent of SSRN-matched citations resolve to published articles, with top destinations in the *Journal of Financial Economics*, *The Journal of Finance*, and the *Review of Financial Studies*. Working papers are a real channel through which frontier research reaches rulemaking, but mostly as an access format rather than a preview of unpublished findings.

The 2024 peak (8,778 citations) does not reflect a handful of outlier rules. The five most citation-heavy rules of 2024—including EPA's light- and heavy-duty vehicle emissions standards, CMS's inpatient payment rule, and the Department of Labor's fiduciary rule—account for 22.6 percent of the year's citations, a *lower* concentration than in typical earlier years (25–37 percent). The 2024 increase is broad-based, consistent with an end-of-administration surge of evidence-intensive rulemaking across many agencies rather than one or two exceptional records.

Finally, the citation counts point to a small group of researchers whose work is unusually embedded in regulation. The most-cited authors at HHS are health services researchers studying Medicare quality, a literature with a direct pipeline from published findings to payment-rule provisions. Whether such citations indicate genuine influence, post-hoc justification, or both is a question the index makes tractable but does not answer; pairing it with measures of regulatory outcomes is a natural next step.

5 Conclusion

We present a comprehensive citation index for federal regulations, covering three decades of rulemaking and linking 85,327 citation instances to 55,520 unique papers by 72,308 unique authors. The data reveal substantial growth in the use of academic evidence in rulemaking—on both the extensive margin (more rules citing research) and the intensive margin (denser citation within rules)—together with strong disciplinary specialization across agencies and a broad evidence base spanning several thousand journals.

The index opens several lines of inquiry. Researchers can study whether regulations that cite more or higher-quality evidence fare better in judicial review, particularly as courts scrutinize agency records more closely after *Loper Bright*. The dataset can inform debates about whether agencies rely on the “best available science” or exhibit selection in what they cite. And by identifying which researchers are most frequently cited in regulations, the index provides a measure of policy impact that complements academic citation counts.

The dataset and replication code will be posted in a public repository; in the interim, both are available from the authors on request.

A Top 100 Cited Sources

Table 6: Top 100 Most-Cited Sources in Federal Regulations, 1995–2025

Rank	**Source**	**Cites**	**Rank**	**Source**	**Cites**
1	JAMA	430	51	Journal of Banking & Finance	63
2	New England Journal of Medicine	399	52	New York Times	62
3	Health Affairs	371	53	Journal of Public Economics	61
4	Journal of Financial Economics	325	54	Journal of Arthroplasty	61
5	The Journal of Finance	320	55	Journal of Food Protection	61
6	Environmental Health Perspectives	267	56	Medical Care Research and Review	60
7	MMWR	245	57	Nature	58
8	Environmental Science & Technology	189	58	Wall Street Journal	57
9	SAE Technical Paper Series	176	59	Clinical Infectious Diseases	56
10	J. American Geriatrics Society	170	60	Journal of Hospital Medicine	54
11	American Economic Review	166	61	Infection Control & Hospital Epi.	53
12	Annals of Internal Medicine	162	62	Critical Care Medicine	50
13	Energy Policy	139	63	J. Policy Analysis and Management	49
14	Medical Care	130	64	Annals of Emergency Medicine	48
15	J. General Internal Medicine	130	65	ATSDR's Toxicological Profiles	47
16	Quarterly Journal of Economics	128	66	Federal Reserve Bulletin	44
17	Review of Financial Studies	125	67	Clinical J. American Society of Nephrology	44

Table 6 continued

Rank	Source	Cites	Rank	Source	Cites
18	Atmospheric Environment	122	68	medRxiv	44
19	JAMA Network Open	121	69	Diabetes Care	44
20	Proc. Natl. Acad. Sci.	114	70	Chest	43
21	Health Services Research	113	71	Am. J. Infection Control	43
22	The Lancet	107	72	Am. J. Preventive Medicine	43
23	American Journal of Public Health	101	73	J. American Med. Directors Assoc.	42
24	Archives of Physical Med. and Rehabilitation	94	74	Journal of Health Economics	42
25	British Medical Journal	93	75	Environmental Pollution	42
26	Journal of Political Economy	93	76	Atmospheric Chemistry and Physics	41
27	Am. J. Respiratory and Critical Care Med.	91	77	Kidney International	41
28	Circulation: Cardiovascular Quality and Outcomes	90	78	Review of Economic Studies	41
29	Am. J. Kidney Diseases	89	79	American Journal of Epidemiology	40
30	Sci.: A J. American Assoc. for the Advancement of Sci.	89	80	Transp. Res. Record: J. Transp. Res. Board	38
31	J. Accounting & Econ.	88	81	Obstetrics & Gynecology	38
32	The Accounting Review	87	82	Environmental Research	38
33	Archives of Internal Medicine	86	83	J. Vascular and Interventional Radiology	38
34	Finance and Econ. Discussion Series	83	84	Psychiatric Services	38
35	Review of Economics and Statistics	80	85	Int. J. Environ. Res. and Public Health	37

Table 6 continued

Rank	Source	Cites	Rank	Source	Cites
36	Blood	80	86	Inhalation Toxicology	37
37	Public Library of Science One	75	87	Stroke	37
38	Pediatrics: Official J. American Academy of Pediatrics	75	88	J. American Society of Nephrology	36
39	J. Financial and Quantitative Analysis	75	89	Emerging Infectious Diseases	36
40	Journal of Economic Perspectives	74	90	Journal of Corporate Finance	35
41	Journal of Accounting Research	71	91	Journal of Vascular Surgery	35
42	Journal of Clinical Oncology	71	92	American Economic Journal	35
43	JAMA Internal Medicine	69	93	Am. J. Clinical Nutrition	35
44	Am. J. Industrial Medicine	68	94	American Journal of Managed Care	35
45	Circulation: J. American Heart Assoc.	68	95	Financial Analysts Journal	34
46	J. Geophysical Res.: Atmospheres	68	96	Am. J. Agricultural Economics	34
47	Management Science	66	97	Cochrane Database of Systematic Reviews	34
48	J. Air & Waste Management Assoc.	66	98	Environment International	34
49	Health Affairs Forefront	65	99	The Journal of Law and Economics	33
50	J. American College of Cardiology	64	100	Brookings Papers on Econ. Activity	33

Notes: Counts are citation instances. The table ranks cited venues as they appear in the standardized data; not all are peer-reviewed journals—

newspapers (*The New York Times*, *Wall Street Journal*), working-paper series (Finance and Economics Discussion Series), preprint servers (medRxiv), and government serials (*Federal Reserve Bulletin*) appear where agencies cite them frequently. Abbreviations follow the scheme of Table 4.

B Data Construction Methodology

This appendix provides a detailed description of the data construction pipeline, from the raw Federal Register corpus through the final citation index. The pipeline consists of six stages: (1) corpus assembly, (2) citation extraction, (3) bibliographic enrichment, (4) journal standardization, (5) author disambiguation and paper deduplication, and (6) working-paper resolution and misattribution correction. Figure 2 provides a schematic overview. The pipeline is deterministic: rebuilding the index from the enriched citation files reproduces the published data exactly.

Stage 1: Corpus Assembly
Federal Register API → 113,712 final rule texts (1995–January 2025)

Stage 2: Citation Extraction
Footnote parsing → Legal citation filtering → Chunking (batches of 50) → Claude Sonnet 4 (Batch API) → Structured JSON

Stage 3: Bibliographic Enrichment
CrossRef API (DOI lookup → title search → author+year fallback)

Stage 4: Journal Standardization
ISSN-based registry from Web of Science indices + JCR + manual fixes

Stage 5: Disambiguation and Deduplication
Paper dedup (DOI → title+year → fuzzy match) → Author disambiguation (Union-Find)

Stage 6: Working-Paper Resolution and Misattribution Correction
SSRN → OpenAlex → published journal; platform and weak-match reversion

Figure 2: Data construction pipeline overview.

B.1 Corpus Assembly

We obtained metadata for every document published in the Federal Register from 1995 through January 2025 via the Federal Register API (`federalregister.gov/api`). For each document classified as a "Final Rule," we downloaded the full text, yielding 113,712 documents. Document metadata includes the title, publication date, document number, issuing agencies, and regulatory identification numbers.

The full-text files follow the typographic conventions of the Federal Register, with numbered footnotes marked by backslash-delimited superscripts (e.g., `\1\` through `\N\`). These footnotes contain the vast majority of citations in regulatory text, including references to academic literature, government reports, legal authorities, and other sources.

B.2 Citation Extraction

Citation extraction proceeds in three sub-steps: footnote identification, legal citation filtering, and large language model (LLM) parsing.

B.2.1 Footnote Identification

We extract numbered footnotes from each document using the Federal Register's typographic convention. Each document's text is segmented at section dividers (lines of ten or more hyphens), and within sections beginning with a numbered footnote marker, we apply a regular expression to capture each footnote's number and content:

```
\\(\d+)\\\s+(.*?)(?=(?:\\\d+\\|\Z))
```

This pattern matches the backslash-delimited footnote number and captures all text up to the next footnote marker or end of section. Footnotes shorter than 20 characters and cross-references (e.g., those beginning with "Ibid.," "Id.," or "See supra") are excluded.

In addition to footnotes, we extract parenthetical citations of the form "Author et al. (Year)" or "(Author, Year)" from the body text, along with roughly 300 characters of surrounding context. We also identify formal reference sections using header patterns (e.g., "References," "Bibliography," "Works Cited") and extract individual entries. These supplementary extractions capture citations that appear outside the footnote apparatus.

B.2.2 Legal Citation Pre-Filtering

Federal Register footnotes frequently cite legal authorities rather than academic sources. To reduce unnecessary API costs, we pre-filter footnotes locally using a battery of regular expressions that match common legal citation formats. Table 7 lists the principal patterns. Footnotes matching any of these patterns are excluded before submission to the language model. This filter removes the majority of footnotes—those citing CFR sections, statutes, court opinions, and other legal materials—while retaining those that potentially reference academic or scientific literature. A limitation is that a footnote containing both a legal citation and an academic citation is dropped whole; the validation sampling in the replication package is designed to quantify this loss.

Table 7: Legal Citation Patterns Used for Pre-Filtering

Citation Type	**Pattern (simplified)**
Code of Federal Regulations	`\d{1,2} CFR \d+`
United States Code	`\d{1,2} U.S.C. \d+`
Federal Reporter (2d, 3d)	`\d+ F.2d \d+`, `\d+ F.3d \d+`
U.S. Reports	`\d+ U.S. \d+`
Supreme Court Reporter	`\d+ S.Ct. \d+`
Public Law	`Pub. L. No. \d+`
Federal Register	`Fed. Reg.`
Statutes at Large	`\d+ Stat. \d+`
Executive Orders	`E.O. \d+`
Court cases	`v. [A-Z]`

B.2.3 LLM-Based Extraction

We use Anthropic's Claude Sonnet 4 (`claude-sonnet-4-20250514`) to classify and parse the remaining footnotes into structured citation records. The model receives a system prompt instructing it to exclude legal citations and extract only academic and scientific references, and a user prompt containing the footnote text. For each citation identified, the model returns a JSON object with the following fields: the original citation text, parsed title, author names, publication year, journal or publisher name, DOI (if present), citation type (journal article, book, report, working paper, conference paper, dissertation, or other), a flag for truncated author lists ("et al."), and any organizational author.

Chunking. Documents with many footnotes are split into consecutive chunks of 50 footnotes each. Each chunk is submitted as a separate API request with a custom identifier encoding both the document number and chunk index. Without chunking, we found that the model would sometimes stop early on documents with hundreds of footnotes, producing incomplete extractions. Setting the maximum output length to 16,384 tokens per chunk and limiting each chunk to 50 footnotes ensures complete processing. During post-processing, results from all chunks belonging to the same document are recombined.

Batch API. All requests are submitted via Anthropic's Message Batches API, which processes requests asynchronously at a 50 percent discount relative to synchronous API calls. The full corpus was processed in nine batches organized by publication-year ranges, with each batch typically completing within 24 hours. Total API cost for the full extraction was approximately $550.

B.3 Bibliographic Enrichment

We enrich extracted citations by querying the CrossRef API (`api.crossref.org`) to obtain DOIs, canonical journal names, ISSNs, volume and page information, and structured author data. We use the "polite pool" (providing a contact email in the `mailto` parameter) for faster rate limits.

B.3.1 Three-Tier Matching Strategy

For each citation, we attempt to match it to a CrossRef record using a cascading strategy:

1. **DOI lookup.** If the extracted citation includes a DOI, we query CrossRef directly by DOI. This is the most reliable matching method and yields an exact match (match score = 1.0).

2. **Title search.** If no DOI is available and the title is at least 10 characters long, we search CrossRef by title (`query.title` parameter, returning up to 3 results). We

compare the extracted title against each returned title using the `SequenceMatcher` algorithm and accept the best match if its similarity ratio exceeds 0.70. Matches with similarity below 0.80 are later discarded when the underlying citation is not a person-authored article (Section B.7).

3. **Author+year search.** If neither DOI nor title matching succeeds, and both an author name and year are available, we search CrossRef by the first author's last name (`query.author` parameter, returning up to 10 results). We accept a match if the publication year matches exactly and the title similarity exceeds 0.50 (a lower threshold reflecting the less precise search).

All CrossRef responses are cached locally, keyed by DOI or normalized title, to avoid redundant API calls across processing runs; transient network errors are cached separately from genuine non-matches so they are retried.

B.3.2 Field Resolution

For each citation, we maintain parallel fields preserving the original LLM extraction (prefixed `claude_`) and the CrossRef match (stored in `crossref_raw`). "Final" fields are resolved by preferring CrossRef data when a match exists and falling back to the LLM extraction otherwise. This dual-track approach allows us to assess enrichment quality and revert to original extractions when CrossRef matching proves unreliable.

B.4 Journal Standardization

Journal names in academic citations appear in many variant forms: full titles, standard abbreviations, non-standard abbreviations, and occasional misspellings. We standardize journal names using an ISSN-based registry constructed from multiple authoritative sources.

B.4.1 Registry Construction

The journal registry ingests the following sources:

1. **Web of Science citation indices.** We load journal lists from the Science Citation Index Expanded (SCIE), Emerging Sources Citation Index (ESCI), and Arts & Humanities Citation Index (AHCI). Each provides the canonical journal title, print and electronic ISSNs, and Web of Science subject categories.

2. **Journal Citation Reports (JCR).** The JCR file provides mappings from standard 20-character abbreviated titles to full journal titles, along with index membership flags. This adds approximately 21,800 abbreviation-to-title mappings.

3. **Bulk abbreviation files.** Two supplementary mapping files (`journals_hash.json` and `entity_hash.json`) contribute abbreviation expansions for journals and organizations accumulated in prior research.

4. **Manual corrections.** A curated file (`journal_fixes.json`, roughly 3,600 entries) records manual mappings for common abbreviations and name variants (e.g., "JAMA" → "Journal of the American Medical Association"). It is loaded last and takes priority over the bulk files, and its mappings are also applied to the canonical values contributed by other sources, so that variant canonical spellings of the same journal are unified.

5. **CrossRef container data.** From the CrossRef responses in our enriched citations, we extract container-title and ISSN pairs, adding associations not already present. CrossRef data never overrides the curated sources; it fills gaps and improves display casing only.

The resulting registry maps roughly 48,800 canonical journal titles and 37,300 ISSNs. The registry is a lookup resource; 5,691 distinct journals actually appear in the citations.

B.4.2 Normalization Algorithm

For each citation's journal name, the lookup proceeds through five steps:

1. Exact match against the lowercase title index.
2. Match after removing a leading article ("The").
3. Match against a "cleaned" index that strips periods, hyphens, commas, ampersands, and whitespace.
4. Cleaned match after removing a leading article.
5. If a match is found and both "The X" and "X" variants exist, unify to the variant with a known ISSN.

B.5 Paper Deduplication

The same paper may be cited in multiple regulations with slightly different title spellings, author orderings, or metadata. We deduplicate papers using a two-pass approach.

B.5.1 Pass 1: Exact Key Grouping

Each citation is assigned a deduplication key using the following hierarchy:

1. **DOI** (most reliable): normalized to lowercase with the `https://doi.org/` prefix removed.
2. **ISSN + volume + first page**: for citations with CrossRef data that includes these fields, catching cases where titles differ slightly.
3. **Normalized title + year**: the title is lowercased, stripped of punctuation, and truncated to 120 characters; combined with the publication year.
4. **First author + year + journal**: used when the title is too short or missing.

5. **Text hash**: a deterministic hash of the original citation text as a last resort.

Citations sharing the same key are grouped together. For each group, we select the citation with the most non-empty fields as the base record, merge author lists across all citations in the group, and record all unique citing documents.

B.5.2 Pass 2: Fuzzy Title Matching

After exact grouping, we perform a second pass to merge groups with nearly identical titles. We block candidate pairs by shared publication year and first author last name (for computational tractability), then compare titles using the `SequenceMatcher` algorithm. Groups whose titles exceed a similarity threshold of 0.90 are merged; chains of merges are resolved before application, and the key remapping is retained so that every citation instance links to a surviving paper record.

B.5.3 Iterative Enrichment

Two additional passes propagate metadata across the corpus:

- **Within-corpus matching**: Papers lacking a DOI or CrossRef match are compared against a "confident pool" of papers that do have CrossRef data. Matches with title similarity above 0.85 (within the same year and first-title-word block) inherit the confident paper's DOI, journal, publisher, and author data, and are merged into its record.

- **Cross-citation propagation**: Papers sharing the same deduplication key have their metadata merged, with non-empty fields from the best-matched source taking precedence.

B.6 Author Disambiguation

Author names in regulatory citations appear in many forms: full names ("Harlan M. Krumholz"), abbreviated names ("H. M. Krumholz"), initials only, last name only ("Krumholz"), and occasionally inverted or misspelled variants. Our disambiguation pipeline resolves these variants into unique author identities using a Union-Find algorithm with corroborating evidence. Throughout, the design errs on the side of *not* merging: a false merge fabricates a prolific author who does not exist, while a false split merely spreads one person's citations over two identities.

B.6.1 Name Normalization

Before disambiguation, all author names are normalized:

- Names are converted to lowercase with standardized "Last, First" format; hyphenated and spaced compound surnames are treated identically ("Joynt-Maddox" ≡ "Joynt Maddox").
- Periods are removed from initials.
- Repeated name segments are detected and removed (e.g., "doty, michelle m doty" → "doty, michelle m").
- Inverted names are detected heuristically: if the putative last name is a short vowel-less string of initials (excepting the genuine surname "Ng") and the putative first name is longer and resembles a real name, the parts are swapped.
- A dictionary of manual corrections (`author_fixes.json`) handles known edge cases, and a curated dictionary of 9,873 entries (`citation_authors_dict.json`) maps raw citation text to full author name lists where the original text was ambiguous.

B.6.2 CrossRef Name Bank

We construct a "name bank" from the structured author fields in CrossRef responses across the entire corpus. For each CrossRef author record containing a family name and given name, we index the full given name by the tuple (last name, first initial). This allows us to resolve abbreviated names when the resolution is unambiguous: if "Krumholz, H." is compatible with exactly one banked name, it is expanded to "Krumholz, Harlan M." When several banked names share the same initial (as with common surnames), no expansion is made. A list of 436 government agency names distinguishes organizational authors from personal names.

B.6.3 Union-Find with Corroborating Evidence

Author name variants are grouped by (last name, first initial) into candidate clusters. Within each cluster, we apply a Union-Find algorithm that merges two variants only when supported by corroborating evidence:

1. **Same paper with name subsumption**: Two variants that appear as authors of the same paper are merged only when one name is an abbreviation-compatible form of the other ("J. Smith" subsumes "John A. Smith"). Same-paper overlap alone is not sufficient—two distinct co-authors can share a surname and initial, and appearing together is, if anything, evidence that they are different people.

2. **Co-author overlap with name subsumption**: Two variants that share at least one co-author across their respective papers, and where one name subsumes the other, are merged.

3. **CrossRef structured name match**: If the CrossRef name bank maps both variants to the same unique full name, they are merged.

4. **Unique first-name stem**: If all full-name variants within a (last name, initial) group

share a single first-name stem (e.g., all “M. Elliott” variants resolve to “Marc”), initial-only variants are merged with the full-name form.

Abbreviation compatibility itself requires unambiguity: a first-name prefix (“chris” for “christopher”) counts as an abbreviation only when the candidate group contains a single first-name stem. This guard matters for dense surname groups—among authors named “Y. Wang,” the given names Yan, Yang, Yanbo, Ying, and Yun all share prefixes, and treating prefixes as abbreviations would chain the entire group into a single spurious identity through Union-Find transitivity.

After the main Union-Find pass, three post-processing passes handle special cases:

- **Compound last names**: Detects misparses of compound surnames (e.g., “Riegg Cellini, Stephanie” vs. “Cellini, Stephanie Riegg”) and merges the variants.
- **Surname changes**: Merges a compound-surname identity with its base-surname identity (e.g., “Joynt Maddox, Karen E.” with “Joynt, Karen E.”) when the first names share a full stem and the two identities share a co-author or a paper.
- **Flipped names**: Detects first/last name swaps (e.g., “Ashish K., Jha”) and corrects the canonical form.

For each cluster, the canonical display name is the CrossRef bank name when it is an unambiguous expansion of a cluster variant, and otherwise the longest variant. Bare last names (e.g., “Smith” with no first name) are merged with a full-name identity only if exactly one canonical author shares a paper with the bare name; the many remaining bare-name identities are excluded from author-level rankings.

This procedure consolidates 24,145 variant names, yielding 72,308 unique author identities.

B.7 Working-Paper Resolution and Misattribution Correction

Automated title matching against a bibliographic database produces characteristic errors, because the database indexes repositories, review services, and platform entries in addition to true sources. We correct three systematic classes; every corrected record retains a flag and the discarded match, so all decisions are auditable in the published data.

B.7.1 SSRN Resolution via OpenAlex

CrossRef sometimes attributes a citation to the "SSRN Electronic Journal" when a copy of the cited work was posted to SSRN. For the 1,075 unique papers so attributed, we query the OpenAlex API (`api.openalex.org`) by title to determine whether a published journal version exists. We verify that (a) the OpenAlex source is a genuine publication venue (repositories and eBook platforms are rejected), (b) the DOI does not carry the SSRN prefix (`10.2139`), (c) the publication year does not predate the cited working paper by more than a year, and (d) author last names overlap where author information is available. Of 1,075 SSRN papers, 500 (47 percent) resolve to published journal articles, updating 770 citation instances; the top destination journals are the *Journal of Financial Economics* (55 papers), *The Journal of Finance* (48), and the *Review of Financial Studies* (21). The remaining 575 papers—typically government reports, policy documents, or law review articles that someone uploaded to SSRN—are reverted to the original extracted citation.

B.7.2 Platform and Database Reversion

CrossRef title matching also returns entries from review databases, abstract services, and publisher platforms in place of true sources: *Choice Reviews Online* (a book-review index), the *PsycEXTRA Dataset* (a gray-literature database), *CFA Digest* (summaries of articles published elsewhere), publisher eBook platform entries, and similar. Citations matched to these containers are reverted to the original extraction.

B.7.3 Weak Title Matches on Gray Literature

An audit of repeated low-similarity matches revealed a third class: title-search matches with similarity below 0.80 where the language model classified the citation as a report, book, or other non-article source (or found an organizational author, or no authors). These matches are systematically unreliable—OMB's Circular A-4 ("Regulatory Analysis") matched a biomedical article titled "Circular RNA: New Regulatory Molecules"; the Small Business Administration's Table of Size Standards matched a fashion dictionary; EPA's greenhouse-gas inventory matched a book chapter on Kazakhstan's emissions. All title-search matches meeting this rule are reverted to the original extraction.

Table 8: Citation Instances Reverted to the Original Extraction, by Reason

Reversion class	**Citations**
SSRN Electronic Journal (no published version found)	952
Platform and database matches:	
Choice Reviews Online	421
PsycEXTRA Dataset	390
CFA Digest	76
Other databases, abstract services, and eBook platforms	273
Weak title matches on gray literature (similarity < 0.80 rule)	3,309
Total	**5,421**

Notes: Counts are citation instances (not unique papers). Reverted records keep the citation with the fields originally extracted from the regulatory text and retain both a reversion flag and the discarded CrossRef match for auditability.

B.8 Output Data Files

The final citation index consists of three CSV files:

- **Citations file** (85,327 rows): One row per citation occurrence, preserving every instance of a citation in a regulation. Each row records the citing regulation (document ID, agencies, publication date), the cited paper (title, authors, year, journal, DOI), the citation type, CrossRef match metadata, and a paper identifier that links to the papers file.

- **Papers file** (55,520 rows): One row per unique paper after deduplication. Each row records the paper's bibliographic information, the number of citing regulations, the set of citing regulations and agencies, and flags for Web of Science index membership.

- **Authors file** (72,308 rows): One row per unique author after disambiguation. Each row records the canonical name, all observed name variants, total papers and citations, the most-cited journal, the citing agencies, and a co-author count.

References

Mia Costa, Bruce A. Desmarais, and John A. Hird. Science use in regulatory impact analysis: The effects of political attention and controversy. *Review of Policy Research*, 33(3):251–269, 2016.

Melissa Dell. Deep learning for economists. *Journal of Economic Literature*, 63(1):5–58, 2025.

Bruce A. Desmarais and John A. Hird. Public policy's bibliography: The use of research in US regulatory impact analyses. *Regulation & Governance*, 8(4):497–510, 2014.

Jerry Ellig and Patrick A. McLaughlin. The quality and use of regulatory analysis in 2008. *Risk Analysis*, 32(5):855–880, 2012.

Cass R. Sunstein. The office of information and regulatory affairs: Myths and realities. *Harvard Law Review*, 126(7):1838–1878, 2013.

Martin Szomszor and Euan Adie. Overton: A bibliometric database of policy document citations. *Quantitative Science Studies*, 3(3):624–650, 2022.

Wendy E. Wagner. The science charade in toxic risk regulation. *Columbia Law Review*, 95 (7):1613–1723, 1995.